

OSI Model

Topics

- The OSI reference model
- Services in the OSI model

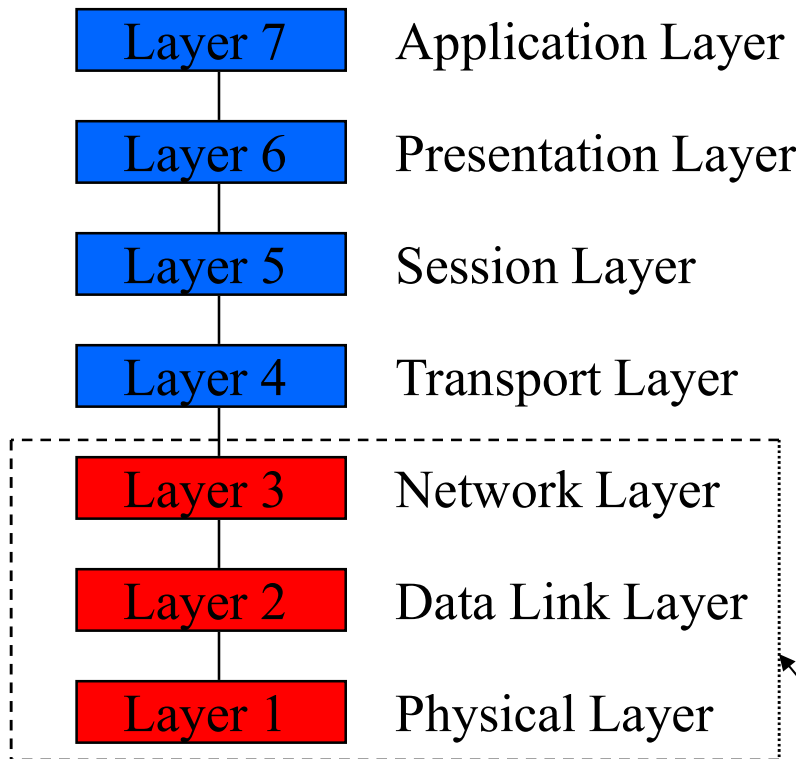
OSI Reference Model

- OSI Reference Model - internationally standardised network architecture.
- OSI = *Open Systems Interconnection*: deals with *open systems*, i.e. systems open for communications with other systems.
- Specified in ISO 7498.
- Model has 7 layers.

- Please Do Not Throw Sausage Pizza Away

- Please → Physical
- Do → Data Link
- Not → Network
- Throw → Transport
- Sausage → Session
- Pizza → Presentation
- Away → Application

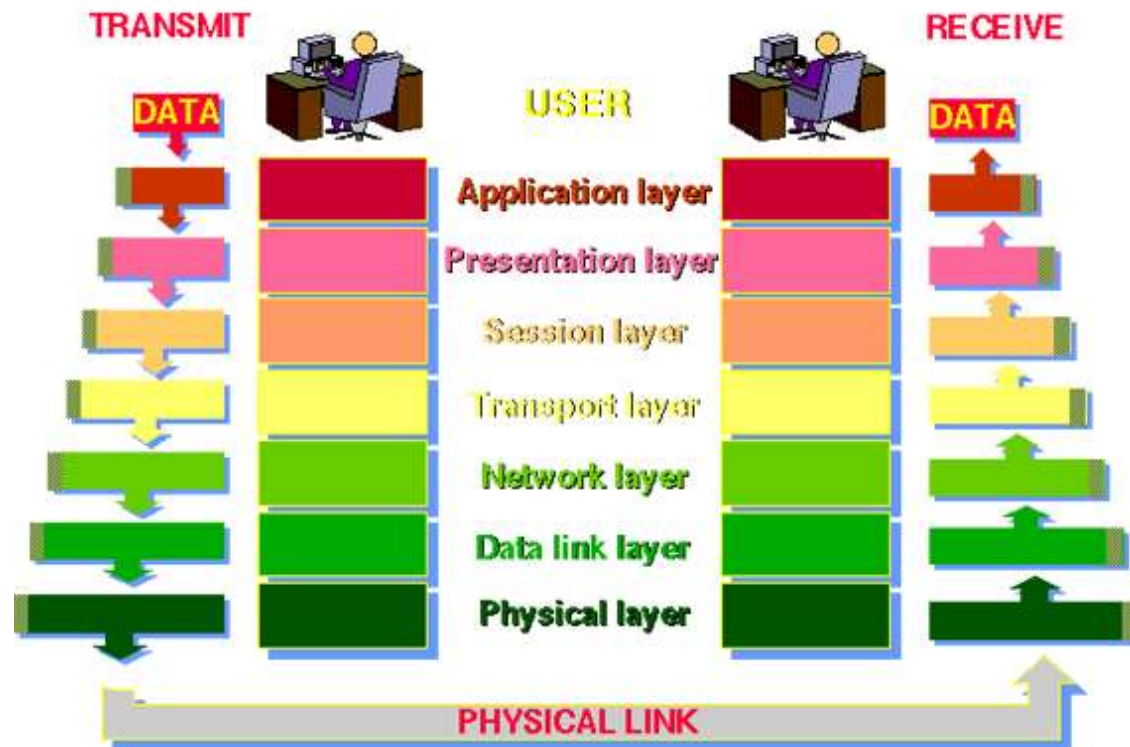
7-Layer OSI Model



- Layers 1-4 relate to communications technology.
- Layers 5-7 relate to user applications.

Communications subnet boundary

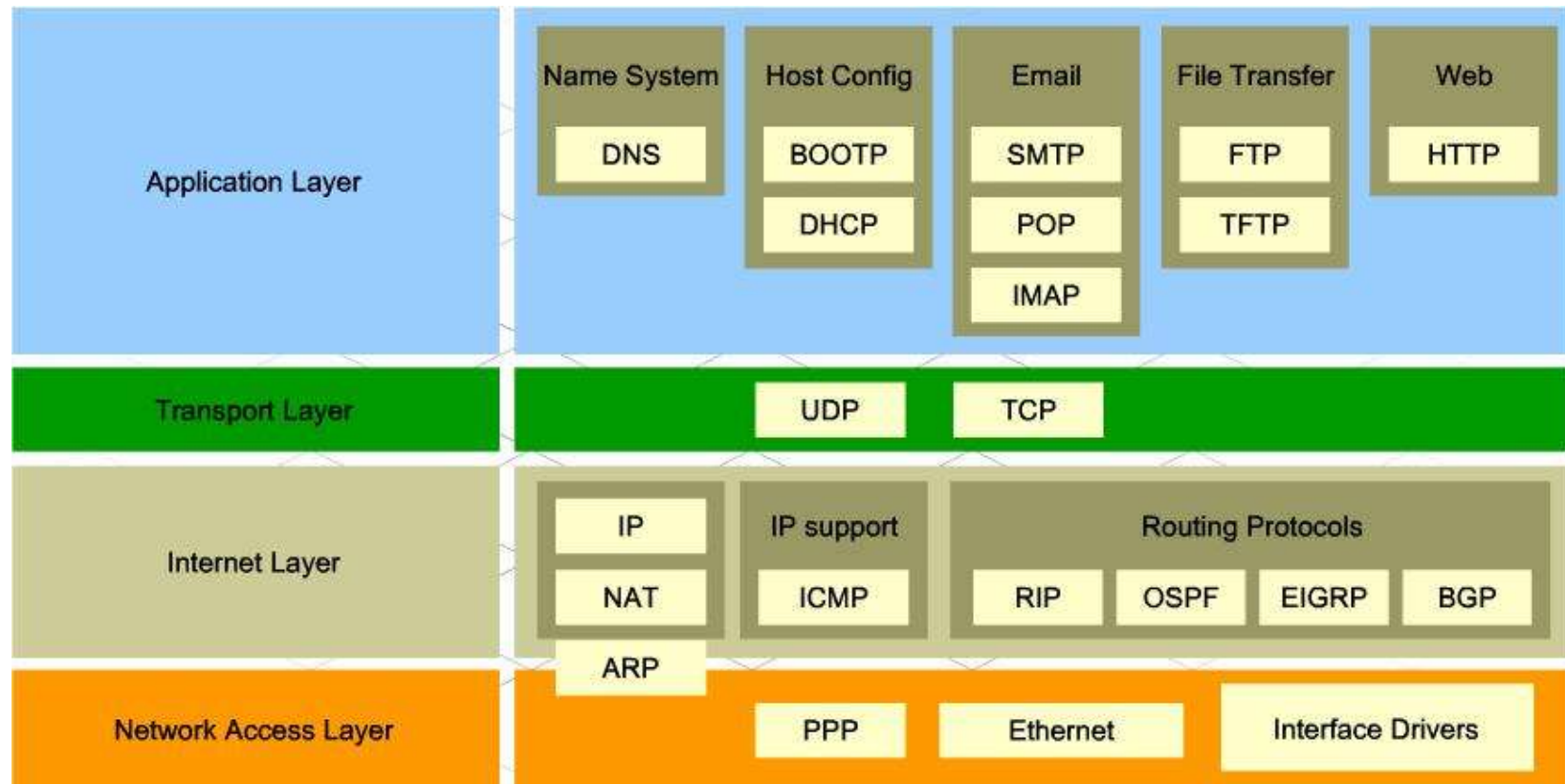
THE 7 LAYERS OF OSI



Layer 7: Application Layer

- Level at which applications access network services.
 - Represents services that directly support software applications for file transfers, database access, and electronic mail etc.

Protocols at Different Layer



Application Protocols Usage

Application	Description
DHCP	Dynamic Host Configuration Protocol assigns IP addresses
DNS	Domain Name System translates website names to IP addresses
HTTP	Hypertext Transfer Protocol used to transfer web pages
NBNS	NetBIOS Name Service translates local host names to IP addresses
SMTP	Simple Mail Transfer Protocol sends email messages
SNMP	Simple Network Management Protocol manages network devices
SNTP	Simple Network Time Protocol provides time of day
Telnet	Bi-directional text communication via a terminal application
TFTP	Trivial File Transfer Protocol used to transfer small amounts of data

Layer 6: Presentation Layer

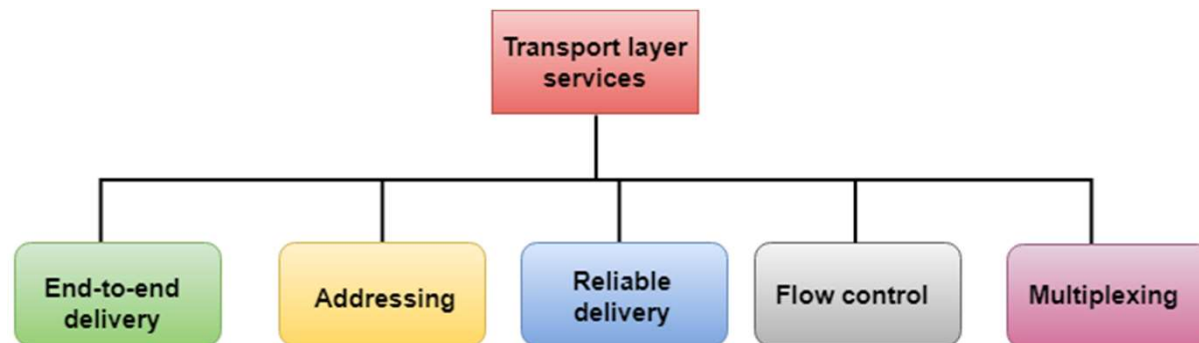
- Related to representation of transmitted data
 - Translates different data representations from the Application layer into uniform standard format
- Providing services for secure efficient data transmission
 - e.g. data encryption, and data compression.

Layer 5: Session Layer

- Allows two applications on different computers to establish, use, and end a session.
 - e.g. file transfer, remote login
- Establishes dialog control
 - Regulates which side transmits, plus when and how long it transmits.
- Performs *token management* and *synchronization*.

Layer 4: Transport Layer

- Manages transmission packets
 - Repackages long messages when necessary into small packets for transmission
 - Reassembles packets in correct order to get the original message.
- Handles error recognition and recovery.
 - Transport layer at receiving acknowledges packet delivery.
 - Resends missing packets



TCP/UDP Header Format

TCP Segment Header Format

Bit #	0	7	8	15	16	23	24	31
0	Source Port				Destination Port			
32	Sequence Number							
64	Acknowledgment Number							
96	Data Offset	Res	Flags		Window Size			
128	Header and Data Checksum				Urgent Pointer			
160...	Options							

UDP Datagram Header Format

Bit #	0	7	8	15	16	23	24	31
0	Source Port				Destination Port			
32	Length				Header and Data Checksum			

TCP/UDP

TCP	UDP
Secure	Unsecure
Connection-Oriented	Connectionless
Slow	Fast
Guaranteed transmission	No Guarantee
Used by critical applications	Used by real-time applications
Packet reorder mechanism	No reorder mechanism
Flow control	No flow control
Error Checking	No Error Checkin
20 Bytes Header	8 Bytes Header
Acknowledgement Mechanism	No Acknowledgement
Three-way handshake (SYN, SYN-ACK, ACK)	No handshake
DNS, HTTP, HTTPS, FTP, SMTP, Telnet, SNMP	DNS, DHCP, TFTP, SNMP, RIP, VOIP

TCP and UDP Application Ports

- Generally Port ranges (0-65535) can be divided into three.
- The zero-port number is reserved and cannot be used, whereas, in UDP, the zero port is not available. These are:

Port Number Range	Port Group
0 to 1023	Well Known (Contact) Ports
1024 to 49151	Registered Ports
49152 to 65535	Private and/or Dynamic Ports

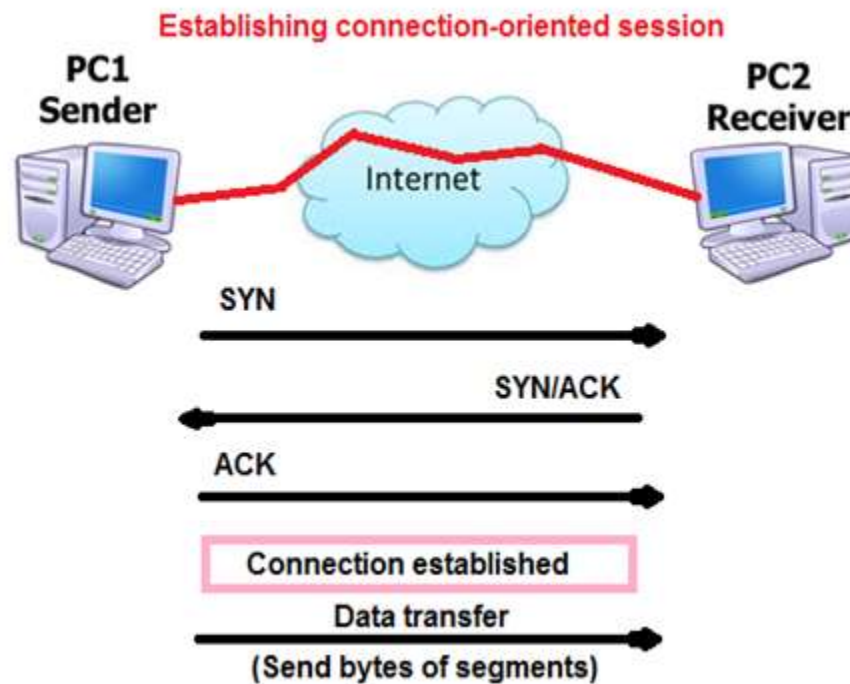


TCP Ports

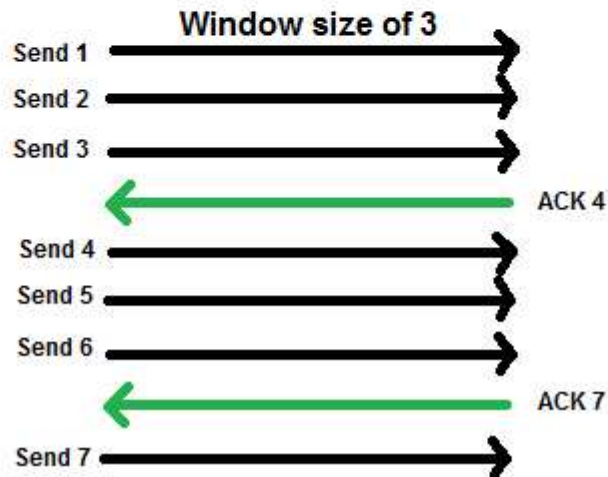
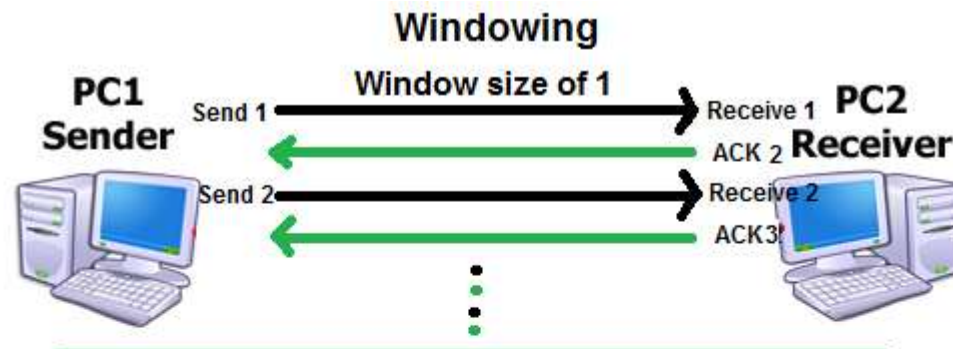


UDP Ports

3 Way Handshake



Windowing



Ack

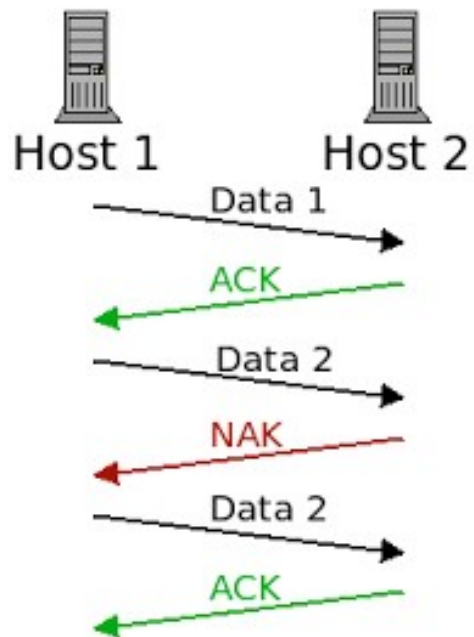
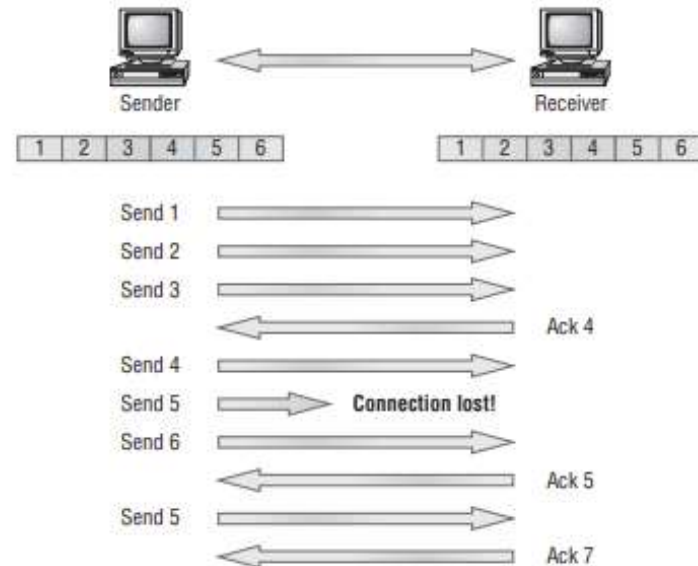
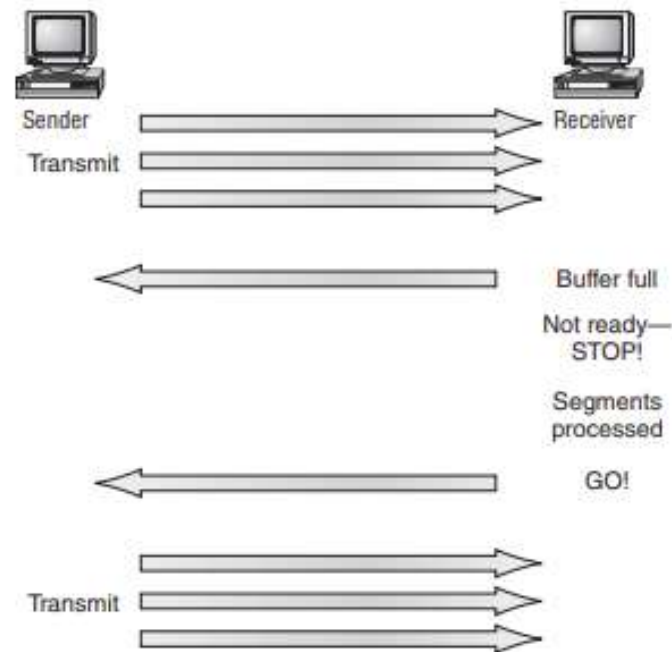


FIGURE 1.12 Transport layer reliable delivery



Flow Control

FIGURE 1.10 Transmitting segments with flow control

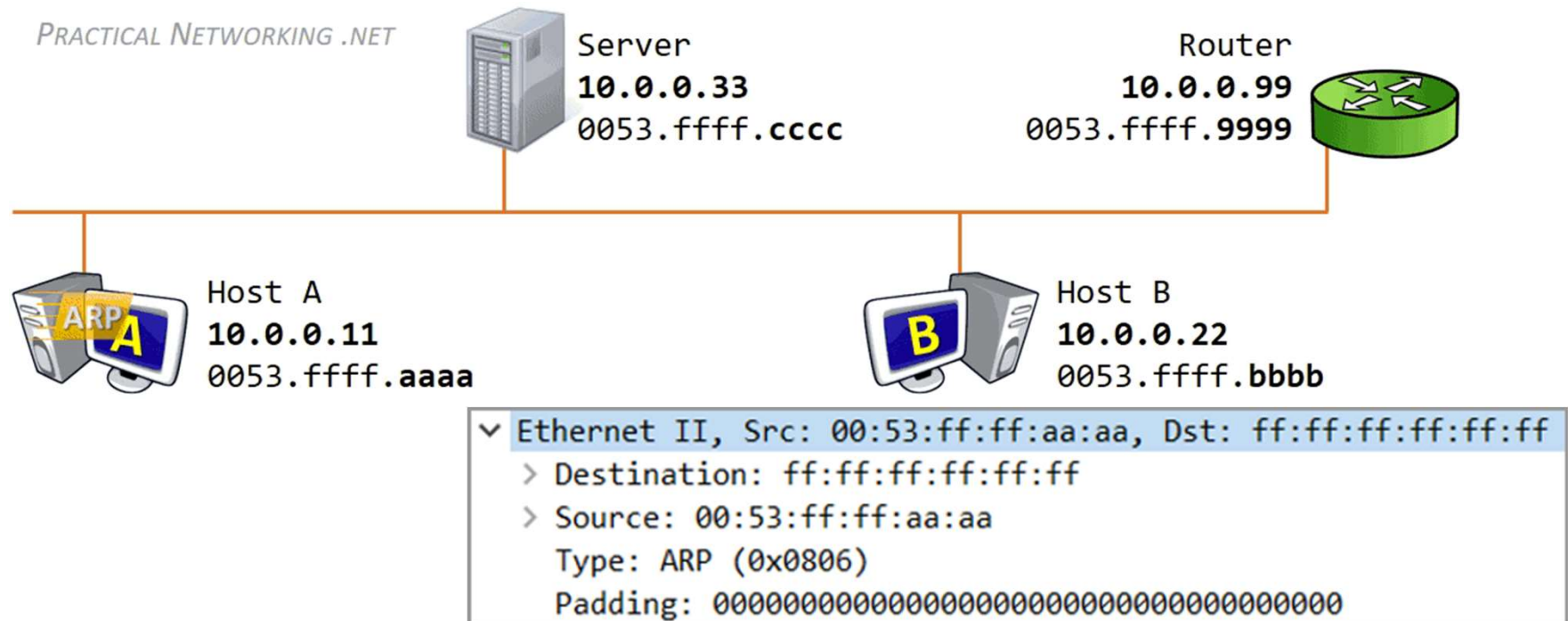


Layer 3: Network Layer

- Manages addressing/routing of data within the subnet
 - Addresses messages and translates logical addresses and names into physical addresses.
 - Determines the route from the source to the destination computer
 - Manages traffic problems, such as switching, routing, and controlling the congestion of data packets.
- Routing can be:
 - Based on static tables
 - determined at start of each session
 - Individually determined for each packet, reflecting the current network load.

ARP

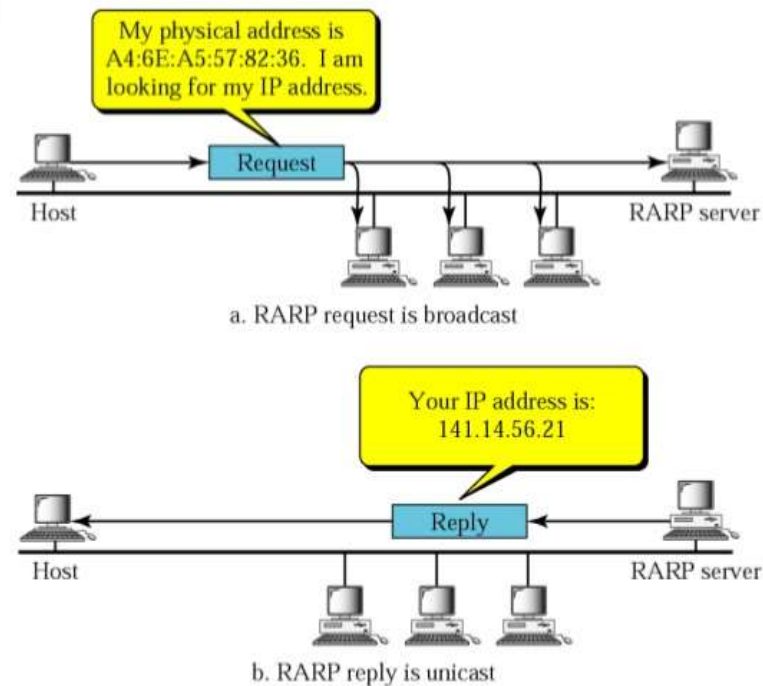
The **Address Resolution Protocol (ARP)** is a communication protocol used for discovering the link layer address, such as a MAC address



RARP

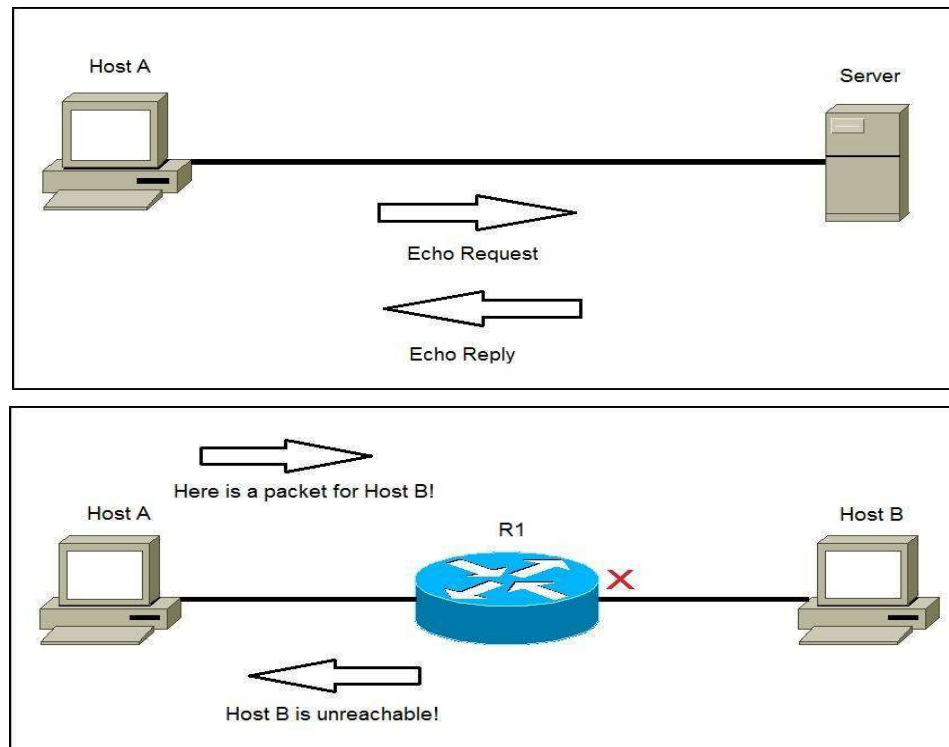
- The **Reverse Address Resolution protocol** is used to translate hardware interface addresses to protocol addresses.

Figure 7.10 RARP operation



ICMP

The **Internet Control Message Protocol** (ICMP) is a supporting protocol in the Internet protocol suite. It is used by network devices, including routers, to send error messages and operational information indicating success or failure when communicating with another IP address, for example, an error is indicated when a requested service is not available or that a host or router could not be reached.



Layer 2: Data Link Layer

- Packages raw bits from the Physical layer into frames (logical, structured packets for data).
- Provides reliable transmission of frames
 - It waits for an acknowledgment from the receiving computer.
 - Retransmits frames for which acknowledgement not received

Layer 1: Physical Layer

- Transmits bits from one computer to another
- Regulates the transmission of a stream of bits over a physical medium.
- Defines how the cable is attached to the network adapter and what transmission technique is used to send data over the cable. Deals with issues like
 - The definition of 0 and 1, e.g. how many volts represents a 1, and how long a bit lasts?
 - Whether the channel is simplex or duplex?
 - How many pins a connector has, and what the function of each pin is?

Internet Protocols vs OSI

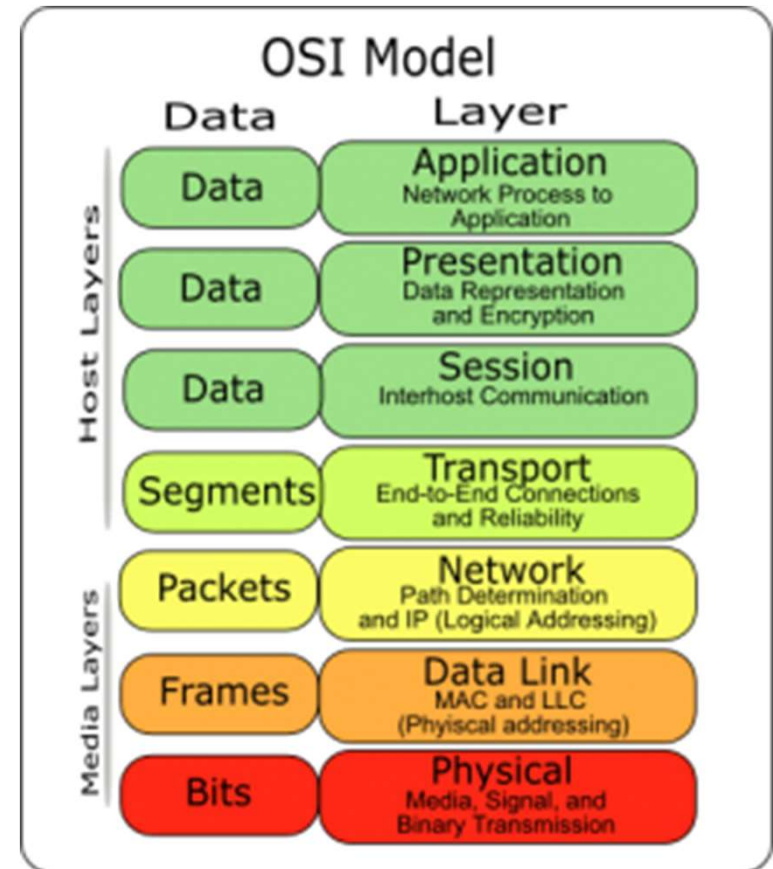
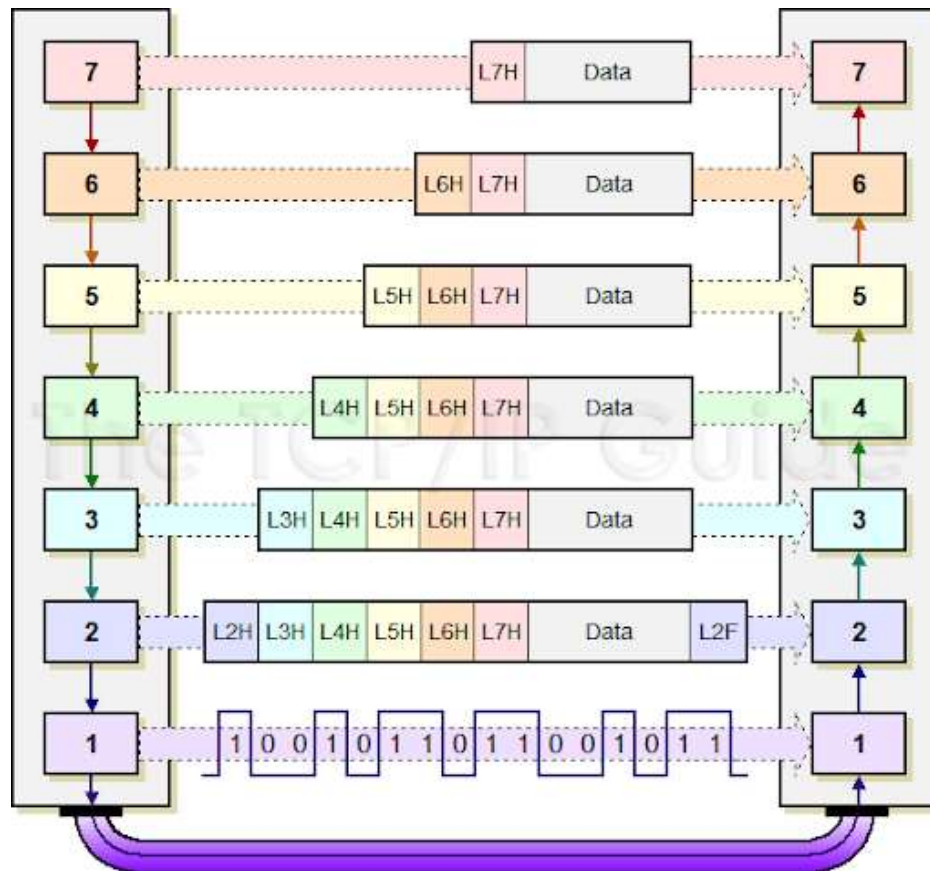
Application		Application
Presentation		
Session		
Transport		TCP
Network		IP
Data Link		Network Interface
Physical		Hardware

- Explicit Presentation and session layers missing in Internet Protocols
- Data Link and Network Layers redesigned

OSI vs TCP/IP Model

	OSI Model	TCP/IP Model	
7	Application	Process/ Application	4
6	Presentation		
5	Session		
4	Transport	Host-to-Host	3
3	Network	Internet	2
2	Data Link	Network Access	1
1	Physical		

Protocol Data Unit(PDU)



Networking Devices Icons



Desktop
Computer



Laptop



Server



LAN Switch



Router



Multilayer Switch



IP Phone



Firewall



Hub



Wireless
Router



Wireless
Access Point



LAN
Media



WAN
Media



Wireless
Media